

EE344: Milestone 1

MON - 10

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Problem Description

Auto-transforming Robot Toy

- To build a physically transforming robotic toy which transforms on user's indication.

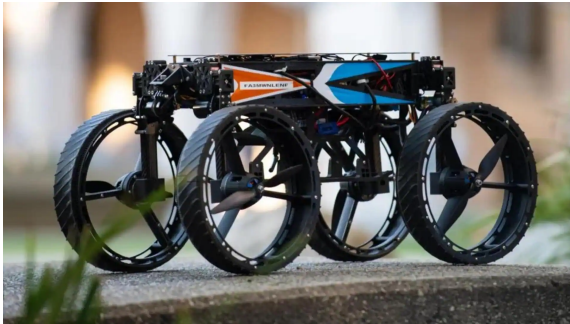


Figure 1: Caltech M4 Morphobot

What is the problem we are solving?

Multi-terrestrial Travel and Transformation

- A ground vehicle cannot move on all terrains and an aerial vehicle has limited flight time.
- Incorporating both modes of travel into singular vehicle form

Possible Applications:

- Drives through debris and flies over obstacles to assist in search and rescue operations.
- Moves stealthily on the ground and takes off for aerial monitoring in restricted areas.
- As entertainment in the form of an autotransforming toy etc.

Proposed Solution

Our idea of an auto-transforming toy is one which can travel on land and in air with a smooth transition (morphosis) between them. We are inspired from the Caltech M4[1]. The toy has 2 distinct modes of operation (ground mode and aerial mode). The morphobot can be controlled manually using a joystick and has flight video feed, for an intuitive user experience. We aim to make it rigid and cost effective to attract target users.

Also other mechanisms (including locking mech.) have been explored for transformation. One of the plausible designs is that of a hybrid mobility robot. [Video Link]

We are going with modified Caltech M4 design, with only drive and drone modes.

How will we achieve the specification?

- We aim to integrate the advantages of ground and aerial modes into a single vehicle.
- The transition consists of rotating the legs by 90° to align parallel to the body, allowing the propellers to function in drone mode.
- To reduce costs, we will operate the bot using a single μ controller. The overall weight has been minimized by eliminating redundant components.
- We have conducted essential calculations to determine the optimal motors for both modes and their transition. The chassis and leg assembly also require significant design considerations.

Design Choices - Caltech M4

Advantages:

- M4 has dedicated wheels, allowing for better ground control, sharp turns, and smooth navigation.
- M4 design allows easier modifications or upgrades (only 2 modes) - cost efficient and smaller design

Disadvantages:

- Bulky size of M4 has to be scaled for practical use
- Servo mechanism for transformations has to be reduced from two hip servos to one (limited modes)

Novelty:

- Hip servos used for the transformation and extension of M4 configurations
- Support structures to balance chassis wt. and individual control of motors and propellers

Design Choices - Hybrid Mobility Robot

Advantages:

- The tank belt allows wheels more mobility in rough terrain
- Transforming mechanism avoids ground impact

Disadvantages

- The exoskeleton design makes it harder to integrate drone components and places more emphasis on mechanical design
- Difficult to modify design according to our requirements (mech.)

Novelty

- Linear actuator for extending propellers to quad-copter mode
- Servo to lower the chassis to ground level to avoid force impact

Design Choices - Microcontroller

- **STM32:**

- We looked at LoRa and STM32 for commands sending but the LoRa was overkill and we decided to go for simpler design
- SpeedyBee had adequate processing and STM was removed to reduce redundant wt.

- **RPi Pico:**

- It has a complete OS which is redundant for our purpose
- Every 10 ms, it executes commands other than flight control

- **ESP32 wifi:**

- The wifi module can be used for commands but inadequate for video feed
- Additional hardware and integration with flight controller

We have decided to work with SpeedyBee as our main as well as flight controller. It has adequate resources.

Principle of Operation

Drive Mode

- The morphobot is capable of ground traversal with a speed range of [0.5 m/s to 2.5 m/s].
- The support structures and chassis design enable, the bot weight balanced by support when wheels leave ground.

Drone mode

- The bot morphs into a drone using high-torque servos, which lift the tire assembly (our calculated max. possible torque of 35 kgcm).
- The drone has been designed (28cm X 28cm dimensions) for indoor flight

Weight Constraints

- The morphobot will weigh upto 3 *kg* taking consideration of all components and mechanical assembly - This was our primary consideration.
- 4S battery preferred over 6S to limit battery weight (234gm).

Power Constraints

- 4S battery of 2200mAh taken appropriate for bldc motor (min. 1500mAh).
- 4S battery chosen over 3S to provide enough power for adequate thrust for 3kg bot (bldc motors)

Technical Details - Drone Mode

- The BLDC motors provide a thrust of 750 gf per motor at 65% throttle.
- The propeller size suitable with the above motor is 6-inch.
- BLDC motor we have chosen is 2550KV with 4S battery (FS 2207.5 Turbo).



(a) Motor for propeller

Throttle (%)	Voltage (V)	Current (A)	Input Power (W)	RPM	Thrust (gf)	Efficiency (gf/w)	MAX
50	15.48	9.86	152.6	16750	494	3.2	62°C
55	15.43	12.1	186.7	17943	576	3.1	
60	15.36	14.67	225.3	18958	665	3.0	
65	15.30	17.35	265.4	19893	754	2.8	
70	15.24	20.22	308.1	21004	835	2.7	
75	15.18	22.8	346.0	21864	900	2.6	
80	15.10	26.19	395.4	22522	987	2.5	
85	15.02	29.22	438.8	23044	1050	2.4	
90	14.94	32.22	481.3	24136	1114	2.3	
95	14.84	35.32	524.0	24679	1170	2.2	
100	14.64	41.17	602.6	25946	1313	2.2	

(b) Rating for motor

Technical Design: Circuit Diagram

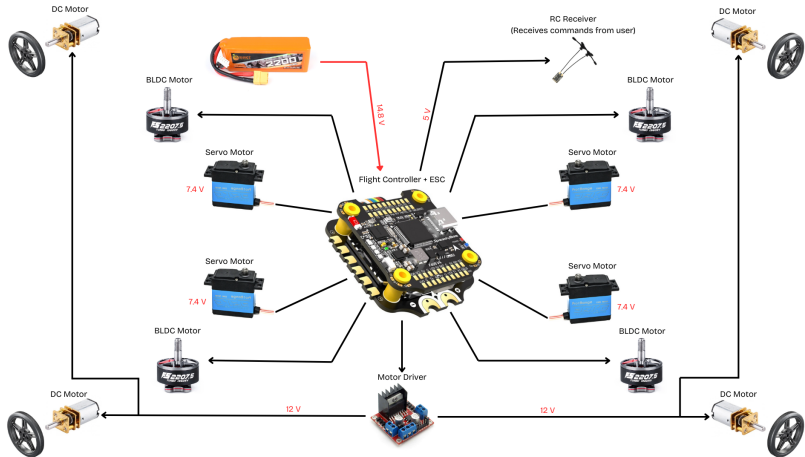


Figure 3: Circuit Diagram

Technical Design: Drive Mode

- Taking propeller diameter 6 inches, wheel diameter 16cm
- Rim thickness - approx. 1-2 cm, Wheel width taken as - approx. 3-4 cm
- High torque and low RPM chosen - minimal wt added



(a) DC motor

Rated Speed (RPM):	10
Rated Voltage (V):	3
Rated Torque (kg-cm):	0.5
Stall Torque (kg-cm):	4
Model No.:	N20-3V-15
Motor Type:	Center Shaft DC Geared Motor
No Load Current (mA)	0.01
Stall current (A):	1
Operating Voltage (VDC):	1.5 ~ 5, 3
No Load Speed (RPM):	15
Gear Ratio:	288:1
Shaft Type:	D Type, Single Side
Motor Body Diameter (mm)	12
Shaft Diameter (mm):	3

(b) Specs for DC motor

Technical Design: Morphing

- Approximate max. torque to be developed is 35 *kgcm* to lift and hold tire assembly
- High torque servo motors to be used
- Possibility of using mechanical locking in the 2 modes



(a) Servo motor

Model No.:	OTS300H
Operating Voltage (VDC)	4 to 6.4
No Load Speed (RPM)	~4286.0V (3.185sec/REV), 5180.4V (3.051sec/REV)
Stall Torque (kg-cm)	28.8 to 26.5
Operating Temperature (°C)	-20 to 60
Motor Supply Current	3.8
Dwell Band Width (Hz)	44
Gear Material	Steel
Rotation Degree(s)	180° Continuous Rotation
Shaft Type	6mm (JST) M3
Cable Length (mm)	30
Command Signal	PWM
Connector	J8
Length (mm)	55
Width (mm)	20
Height (mm)	43
Weight (g)	70
Shipping Weight	0.03 kg
Shipping Dimensions	6 x 5 x 4 cm

(b) Specs for Servo motor

- The chassis would be of the dimensions of $28\text{ cm} \times 8\text{ cm}$.
- In drone mode, entire structure stretches out to $28\text{ cm} \times 28\text{ cm}$.
- Height in the rover mode will be 18 cm .
- Structure should be made of tough but light material. We will explore various options like sheet metal, carbon fibre, etc.
- Additional supports at the bottom of the chassis for smooth transitions in between modes.



(a) Radiomaster Pocket Radio Controller



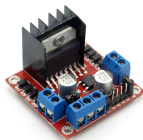
(b) Radiomaster RP3 receiver

Technical Design: Control

- We are planning to use ready-made flight controller. This would ensure that our controller is compatible with the software (eg: **Ardupilot**).
- SpeedyBee has 4 pins to be used for controlling propeller motors and 4 other pins for drive control.
- Additional dedicated pin for servo control for transforming

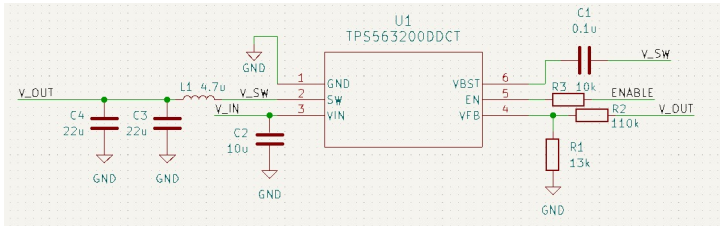


(a) SpeedyBee
F405 V4 BLS 55A
FC & ESC Stack



(b) L298N Motor
Driver

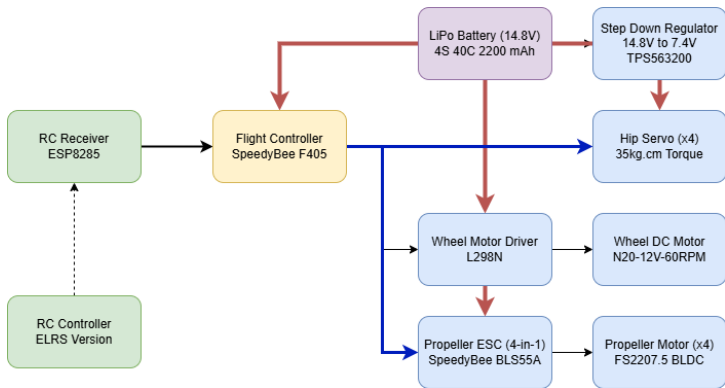
Technical Design: Power Circuit



(a) Step-Down Regulator Circuit

- LiPo Battery (4S 40C 2200mAh)
- Step-Down Switching Regulator (14.8V to 7.4V)

Technical Design: Block Diagram



(a) Step-Down Regulator Circuit

Alternative Options: Communication

- **Zigbee + STM:** This was suitable to send commands with a suitable range and as the main controller. This is a backup, as it is redundant. SpeedyBee provides the same capabilities.
- **ESP32 (Wifi):** A wifi module fits the indoor usage of morphobot. There is also an option of ESPcam for video feed transfer over wifi. We are not using this to maintain minimal weight.
- **FPV VTX and RTX:** This is what we think would be ideal for stable video feed. Since, FPV systems are generally used for drones.

Gantt Chart

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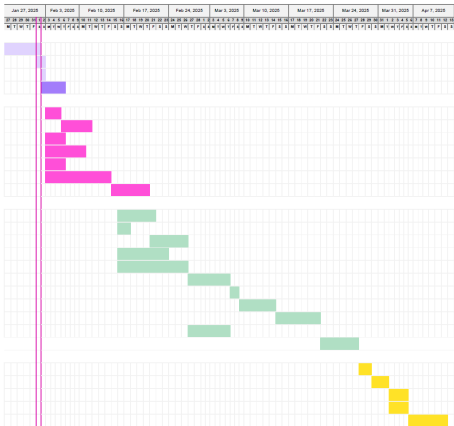
Neel, Shikhi, Saumya, Arin, Harshil

Project start: **Mon, 1/27/2025**

Display week: **1**

TASK	ASSIGNED TO	PROGRESS	START	END
Design Review and BOM				
Create BOM	All	100%	1/27/25	2/1/25
Create Gantt Chart	All	100%	2/1/25	2/2/25
Create Video Presentation	All	100%	2/2/25	2/2/25
Milestone I Evaluation	All	0%	2/2/25	2/2/25
Schematics & Preliminary Analysis				
Choose Flight Controller Firmware	Neel - Shikhi	0%	2/3/25	2/5/25
Go through firmware documentation	Shikhi - Neel	0%	2/5/25	2/11/25
Identify feasible latching mechanisms	Saumya - Neel	0%	2/3/25	2/5/25
Initial Chassis design	Arin - Saumya - Harshil	0%	2/3/25	2/10/25
Develop Power Distribution Board	Harshil	0%	2/3/25	2/5/25
Initial Tite Assembly Design	Saumya - Arin	0%	2/3/25	2/14/25
Milestone II Evaluation	All	0%	2/15/25	2/22/25
Layout, CAD & Initial Testing				
Motor integration & testing for all 3 modes	Neel - Shikhi	0%	2/16/25	2/21/25
Communication Setup	Shikhi	0%	2/16/25	2/17/25
Initial PCB design	Harshil	0%	2/21/25	2/26/25
Finalize Chassis Design	Arin - Saumya	0%	2/16/25	2/23/25
Finalize Tite Assembly Design	Saumya - Arin	0%	2/16/25	2/26/25
First prototype manufacturing	Arin - Harshil	0%	2/27/25	3/2/25
First prototype assembly	Harshil - Shikhi	0%	3/7/25	3/8/25
Transformation PuC	Saumya - Arin	0%	3/9/25	3/14/25
PuG Testing - I	Shikhi - Arin	0%	3/15/25	3/21/25
PCB Assembly and Testing	Harshil	0%	2/27/25	3/5/25
Milestone III Evaluation	All	0%	3/22/25	3/27/25
Final Testing & Evaluation				
Finalize CAD	Arin - Saumya	0%	3/28/25	3/29/25
Final prototype manufacturing & assembly	Saumya - Harshil	0%	3/30/25	4/1/25
Software implementation on final prototype	Neel - Shikhi	0%	4/2/25	4/5/25
PuG Testing - II	Neel - Saumya	0%	4/2/25	4/5/25
End Report and Final Presentation	All	0%	4/6/25	4/12/25

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Bill of Materials

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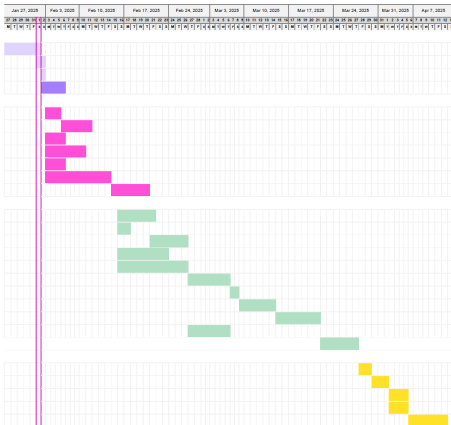
Neel, Shikhi, Saumya, Arin, Harshil

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Create Video Presentation	All	100%	2/2/25	2/2/25
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First prototype manufacturing	Arin - Harshil	0%	2/27/25	3/2/25
First prototype assembly	Harshil - Shikhi	0%	3/7/25	3/8/25
Transformation PoC	Saumya - Arin	0%	3/9/25	3/14/25
PoC Testing - I	Shikhi - Arin	0%	3/15/25	3/21/25
PCB Assembly and Testing	Harshil	0%	2/27/25	3/5/25
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Final Testing & Evaluation				
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PoC Testing - II	Neel - Saumya	0%	4/2/25	4/5/25
End Report and Final Presentation	All	0%	4/6/25	4/12/25

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Risks and Mitigation strategies

- **Weight:** The bot must be lightweight. We have eliminated redundant components and sought lighter alternatives wherever possible.
- **Morphing:** The transition between modes should be smooth, especially from aerial to ground mode. We will perform multiple iterations and simulations.
- **Control:** Operating both modes using a single μ controller increases the complexity of coding, testing, and debugging.
- **Rigidity:** Since toys often endure rough handling, the design/material chosen must be sturdy enough.
- **High Current Flow:** Motor stalling may result in excessive current. We are considering current limiter and other protection mechanisms to prevent this.

To create an **Automated Multi-Modal Mobile Morphobot**

- **Automated:** The robot can transform from ground mode to aerial mode and vice-versa without manual intervention.
- **Multi-modal:** Capable of multiple modes of operation: Ground and Aerial, with scope for more.
- **Morphobot:** The robot morphs i.e. it transforms from one mode to another without using extra hardware.

Thank you!